Chapter 2

Consequences of the Axioms

What is truly fascinating about the three little axioms for a group—associativity, identity and inverses—is that together they imply so much. In this chapter we will look at a number of different immediate consequences of these axioms. We will refer to a consequence of these axioms as a "Lemma", a "Theorem" or a "Corollary". As you prove each of the following consequences, specify each use of one of the axioms.

We begin with a fact that does not even depend on the group axioms. Is it possible for a binary system to have two different identity elements? Try, for example, to fill in a table for a binary operation $*$ on a 3–element set $\{e_1, e_2, a\}$ in which both e_1 and e_2 are identity elements. If you have difficulty, look at exactly what is causing the difficulty and then prove the following theorem.

Theorem 7. *A binary system* **G** *with operation* $*$ *can have at most one identity element.*

Now try to fill in a table for a binary operation $*$ on the 4–element set $\{e, a, b, c\}$ in which e is the (unique) identity and b and c are both inverses for a. This time you should have no trouble, which tells you that any proof of the following theorem will necessarily have to make use the group axioms.

Theorem 8. *Let* **G** *be a group with identity e, and let a be an element of* **G** *with inverse b. If* $c \in G$ *and either* $a * c = e$ *or* $c * a = e$*, then* $c = b$*. In particular, a has only one inverse.*

Since the inverse of a group element a is unique, we will normally denote it by "a^{-1}". In this notation, Theorem 8 says that, if we already know that we are in a group, then we can show that b is the inverse of a by verifying either that $a * b = e$ or $b * a = e$; we don't need to verify both.

Corollary 9. *If* **G** *is a group with identity e and* $a, b \in$ **G***, then*

(i) $(a^{-1})^{-1} = a$

(ii) $(a * b)^{-1} = b^{-1} * a^{-1}$.

Let **G** be a binary system. We say the $e \in$ **G** is a **right identity** if $a * e = a$ for all $a \in$ **G**. An element $b \in$ **G** is a **right inverse** for a (with respect to e) if $a * b = e$. An element $f \in$ **G** is an **idempotent** if $f * f = f$.

Theorem 10. *Let* **G** *be an associative binary system with a right identity e such that each element has a right inverse. Then* **G** *is a group.*

[Hint: You need to show that e is also a left identity and that a right inverse is also a left inverse. Start by showing that e is the only idempotent in $\mathbf{G}$.]

The examples of groups that we have computed will give us a laboratory to test hypotheses that might be true of groups in general. Notice that the two groups $\mathbf{K}_4$ and $\mathbf{P}_{abc}$ have the unusual property that each element is its own inverse. These two are among the commutative groups, but not every commutative group (for example $\mathbf{Z}_{12}$) has this property.

Corollary 11. *Let* $\mathbf{G}$ *be a group in which every element is its own inverse. Then* $\mathbf{G}$ *is commutative.*

Algebra in general grew out of the historical need to solve equations, and this need led specifically to the development of group theory where certain kinds of equations can always be solved.

Theorem 12. *If* $\mathbf{G}$ *is a group,* $a,b \in \mathbf{G}$*, then the equations* $a*x = b$ *and* $y*a = b$ *each have a unique solution, namely,* $x = a^{-1}*b$*, and* $y = b*a^{-1}$.

Cancellation is a technique that is often useful for solving equations, but it must be used judiciously. For example, in high school algebra we cannot conclude from $ab = ac$ that $b = c$ without first checking that $a \neq 0$. In general, for a binary operation $*$, we say that

$*$ is **left cancellative** if $a*b = a*c$ implies $b = c$, and

$*$ is **right cancellative** if $b*a = c*a$ implies $b = c$

for every choice of a,b,c.

Problem 13. *Give examples to show that multiplication of* 2×2 *matrices in general and composition of functions in general are neither left nor right cancellative.*

Theorem 14. *Every group is both left and right cancellative.*

If we examine the group tables that we constructed in Chapter 1, we notice something special about the rows and the columns. In each case, each row and each column is a permutation of the elements of the group. The rows and columns are precisely the functions from the group into itself obtained by left and right multiplication by a single element. What we have observed in these examples is true for all groups.

Theorem 15. *If* $\mathbf{G}$ *is a group and* $a \in G$*, then the two functions*

$$\ell_a(x) := a*x \quad \text{and} \quad r_a(x) := x*a$$

are both permutations of G.

An essential goal of group theory is to discover all possible groups of each finite size. The theory we will develop later will help to do this, but we can already address a simple case. The following theorem can be proven from the axioms, but you might be able to apply what you have learned to give a simpler proof.

Theorem 16. *In the following sense, there is essentially only one group with three elements.*

(i) *There is at most one way to fill in a table for the set* $\{e,a,b\}$*, using* e *as the identity, which could possibly be a group.*

(ii) *The table of part* (i) *is in fact a group.* (*You will need several cases to establish associativity.*)

Associativity in groups allows us to unambiguously write down the product $a*b*c$ where a,b,c are elements of a group **G**. What about longer products such as $a*b*c*d*f$? This expression has quite a number of possible interpretations, for example, $((a*b)*(c*d))*f$ and $a*((b*c)*(d*f))$. Do these both represent the same element? They do, but this fact requires four applications of the associative property to prove:

$$\begin{aligned}((a*b)*(c*d))*f &= (a*b)*((c*d)*f) = (a*b)*(c*(d*f))\\ &= a*(b*(c*(d*f))) = a*((b*c)*(d*f)).\end{aligned}$$

If we could devise a similar argument for every pair of 5–element products, we would then be able to unambiguously write $a*b*c*d*f$.

More generally, is it possible to prove that the product $a_1a_2a_3\dots a_n$ is unambiguous? This task sounds rather daunting, as the number of possible products will grow rapidly as the number of factors extends beyond five. It turns out that it is nevertheless true, and that mathematical induction provides a simple proof. We first define the **left associated product** of $a_1,a_2,a_3,\dots,a_n$ to be

$$(\dots((a_1*a_2)*a_3)*\dots)*a_n.$$

Theorem 17. *Let* **G** *be an associative binary system, and choose elements* $a_1,a_2,a_3,\dots,a_n \in$ **G**. *Then every product of the elements* $a_1,a_2,a_3,\dots,a_n$, *in that order, is equal to the left associated product. In particular, we can unambiguously write down the product* $a_1*a_2*a_3*\cdots*a_n$.

Real number exponents satisfy two important rules:

$$a^m a^n = a^{m+n} \quad \text{and} \quad (a^m)^n = a^{mn}$$

where m and n are positive integers. If a is an element of a group **G** and $n \in \mathbb{Z}^+$ (the set of positive integers), we recursively define the **exponent** a^n as

$$a^1 := a \quad \text{and} \quad a^{n+1} := a^n * a.$$

Applying the above associativity theorem, we see that $a^n := a*a*\cdots*a$, the product of n a's. It is easy to check that the same rules of exponents hold with this definition.

We would like to extend the definition of exponents to give a meaning to "a^0" and "a^{-n}" in such a way that the rules of exponents will continue to hold. There is only one way that we could possibly do this. If $a^0*a^1 = a^{0+1}$, then we have $a^0*a = a$ so that, by Theorem 4, we must define

$$a^0 := e, \quad \text{the identity.}$$

Similarly, if $a^{-n}*a^n = a^{-n+n}$, then we have $a^{-n}*a^n = e$ so that, by Theorem 2, we have no choice but to define

$$a^{-n} := (a^n)^{-1}, \quad \text{the inverse of } a^n.$$

Lemma 18. *If* $\mathbf{G} = \langle G,*\rangle$ *is a group,* $a \in G$ *and* $n \in \mathbb{Z}^+$, *then* $a^{-n} = (a^{-1})^n$.

The proof of the following theorem requires examining a number of cases.

Theorem 19. *Let* $\mathbf{G} = \langle G,*\rangle$ *be a group,* $a \in G$, *and* $m,n \in \mathbf{Z}$. *Then*

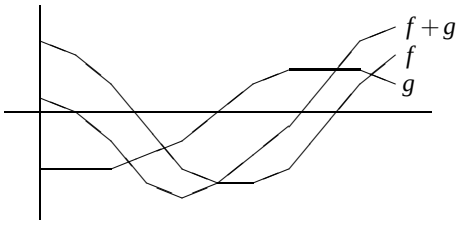
(i) $a^m * a^n = a^{m+n}$ *and*

(ii) $(a^m)^n = a^{mn}$.

For another familiar source of groups, recall that the axioms for a linear space require that vector addition be commutative and associative, and that there be a zero vector which is an additive identity. The other axioms imply that, for each point P of a linear space, the point $(-1)P$ is an additive inverse of P. Thus every linear space is a commutative group under addition.

The group operation in a vector space is normally denoted by the addition symbol +. For example, the vector space of all functions from the real numbers into the real numbers forms a commutative group $\mathbf{F}[\mathbf{R},\mathbf{R}]$ where the sum of functions f and g is defined by

$$(f+g)(x) = f(x)+g(x).$$



When we refer to a commutative group whose operation is denoted by +, such as the additive group of $\mathbf{R}$ of real numbers, the additive group of n-by-m matrices, or the additive group $\mathbf{F}[\mathbf{R},\mathbf{R}]$, it is rather awkward to use the multiplicative notation that we have been using for general groups. Instead, we introduce the **additive notation** for groups as follows.

	Multiplicative	Additive
product; sum	$a*b$ or ab	$a+b$
identity	$ea = ae = a$	$0+a = a+0 = a$
inverse	a^{-1}	$-a$
exponent	a^n	na
quotient; difference	ab^{-1}	$a-b$

In additive notation, for example, Theorem 19 says that, for all $a,b \in \mathbf{G}$ and $m,n \in \mathbf{Z}$, we have

(i) $ma+na = (m+n)a$

(ii) $n(ma) = (nm)a$.

While this may look at first glance like the distributive and associative properties, they are in fact the properties of exponents written in additive notation.